

ADDITIONAL EXPECTED JUDGE QUESTIONS

Real-Time AI + Blockchain Degree Fraud Detection System

Advanced Preparation Guide | BBDU Team

DEEP TECHNICAL QUESTIONS

Q1. OpenCV mein exactly kya detect karta hai tumhara AI?

Specifically teen cheezein — Font inconsistency detection (genuine certificates mein font uniform hoti hai), Edge artifact analysis (tampering se edges blur ya jagged ho jaati hain), aur Metadata anomaly detection (PDF creation date, software used, modification history). Teeno combine hokar Trust Score calculate karte hain.

Q2. Hash collision ka kya — do alag documents ka same hash aa sakta hai?

SHA-256 ke saath collision probability practically zero hai — 2 ki power 256 possibilities hain. Aaj tak koi proven SHA-256 collision exist nahi karta. Yeh industry standard hai — banks aur governments bhi yahi use karte hain.

Q3. Agar AI model ko fool kiya jaye — adversarial attack?

Valid concern hai. Isliye hum sirf AI pe rely nahi karte — blockchain hash verification primary layer hai. AI secondary layer hai. Dono simultaneously fail hona practically impossible hai — yeh dual-layer architecture ka asli faida hai.

Q4. Model kaise train kiya — dataset kahan se liya?

Humne synthetic dataset generate kiya — genuine certificates pe deliberate tampering karke. OpenCV se augmentation ki — brightness, rotation, compression artifacts add kiye. Real-world fake certificate samples publicly available research papers se reference kiye.

Q5. Smart contract mein upgrade mechanism hai kya?

Haan, humne Proxy pattern consider kiya hai future ke liye. Abhi simple upgradeable architecture hai jahan owner address nayi contract deploy karke migrate kar sakta hai without losing historical data.

INNOVATION QUESTIONS

Q6. Trust Score 0-100 kaise calculate hota hai exactly?

Weighted formula hai — Blockchain verification 60% weight, AI visual analysis 30% weight, aur Metadata consistency 10% weight. Agar blockchain match hai lekin AI suspicious patterns detect kare, score 60-70 range mein aayega — flagging manual review.

Q7. Yeh decentralized hai ya centralized — kaun control karta hai system?

Hybrid approach hai. Blockchain layer fully decentralized hai — koi single entity control nahi kar sakti. Lekin web dashboard aur AI module centralized hain abhi. Long-term vision hai fully decentralized DAO-based governance model.

Q8. Kya tumhara system real-time hai ya batch processing?

Real-time hai — single certificate verification 3-5 seconds mein complete hoti hai. Blockchain query ~1-2 sec, AI analysis ~1-2 sec, result display instant. Bulk verification ke liye queue-based batch processing bhi support karta hai.

COMPETITION & MARKET QUESTIONS

Q9. Tumhare competitors kaun hain — directly?

Direct competitors mein Accredible, Blockcerts, aur India mein DigiLocker hain. Lekin inme se koi AI visual forgery detection nahi karta — sirf blockchain registry hai. Hamara hybrid approach unique differentiator hai.

Q10. Agar koi bada company — Google ya Microsoft — yahi feature launch kar de?

Yeh actually validation hai ki problem real hai. Lekin India-specific credential ecosystem, regional university databases, aur local compliance — DPDP Act — ke liye specialized solution chahiye. Big tech generic solutions banate hain, hum India-first approach le rahe hain.

Q11. Pilot test kiya kya kisi real company ya university ke saath?

Abhi hackathon prototype stage pe hain. Lekin BBDU ke saath pilot discussion planned hai. Interested companies ke liye free pilot offer kar rahe hain first 3 months — feedback loop build karne ke liye.

LEGAL & COMPLIANCE QUESTIONS

Q12. Kya yeh legally admissible proof hai agar fraud pakda jaye?

Blockchain records immutable aur timestamped hote hain — yeh strong digital evidence hai. India mein IT Act 2000 ke under digital records legally valid hain. Actual legal proceedings ke liye traditional verification bhi parallel chalti rahegi initially.

Q13. Universities ki permission chahiye hogi data use karne ke liye?

Haan, aur yeh hamara go-to-market strategy ka core part hai. Universities ke saath MOU sign karenge — unhe free certificate issuance tool denge, humhe verified database milegi. Win-win model hai.

TOUGH / TRICK QUESTIONS

Q14. Agar main tumhare system ko aaj hi hack karna chahoon toh kahan se start karoon?

Sabse weak point abhi centralized web server hai — traditional web vulnerabilities. Isliye production mein WAF, rate limiting, aur encrypted API endpoints planned hain. Blockchain layer inherently secure hai — wahan attack surface minimal hai.

Q15. 3 saal mein yeh project band ho jayega — kyun nahi?

Degree fraud problem grow ho rahi hai, shrink nahi. Remote hiring increase se international credential verification ki demand aur badh rahi hai. Aur ek baar universities aur companies onboard ho jayein — switching cost high hoga. Network effect protect karega.

AGAR ANSWER NA PATA HO — KYA BOLNA

Situation	Kya Bolna
Technical depth nahi pata	"Great question — exact detail refine kar rahe hain, but approach yeh hai..."
Data / stats nahi pata	"Exact numbers research kar rahe hain, industry estimate ke hisaab se..."
Out of scope question	"Yeh v2.0 roadmap mein hai, abhi MVP focus pe hain"
Judge disagree kare	"Aapka point valid hai, isliye humne [X] as backup rakha hai"

Confidence hi tumhara sabse bada weapon hai — BBDU ka naam roshan karo! ■